

Relaxation and Odd-Even Effect in Fermi Liquids

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Outline

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Introduction

- k -space
- Fermi surface

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Boltzmann equation

- Linearized collision operator

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Odd-Even effect

1

Introduction

1.1

***k*-space**

k -space

- Analogous to momentum space
- Each point $\mathbf{k}$ in k -space corresponds to a wave $e^{i\mathbf{k}}$
- **Fourier transform** converts between k -space to real space

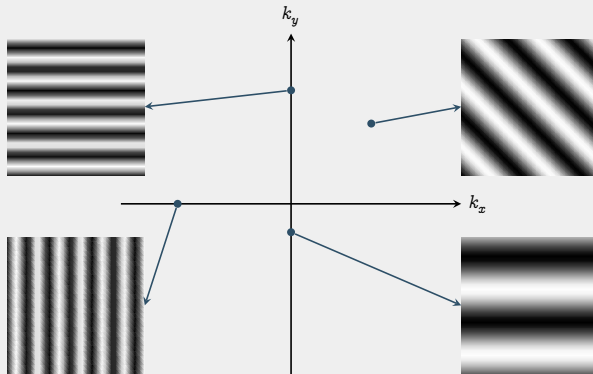


Figure 1: REPRESENTATIVE WAVES OF k -SPACE

- Distance from origin $\propto$ Wave frequency
- Direction on k -space $\implies$ Wave direction

1.2

Fermi surface

2

Boltzmann equation

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{r}} f + \mathbf{F} \cdot \nabla_{\mathbf{v}} f = \left(\frac{\partial f}{\partial t} \right)_{\text{coll}}$$

$$\left(\frac{\partial f}{\partial t}\right)_{\text{coll}} \approx \int \dots \int d^3\mathbf{p}_2 d^3\mathbf{p}'_1 d^3\mathbf{p}'_2 W(\mathbf{p}'_1, \mathbf{p}'_2 | \mathbf{p}, \mathbf{p}_2) [f_2(\mathbf{r}, \mathbf{r}, \mathbf{p}'_1, \mathbf{p}'_2) - f_2(\mathbf{r}, \mathbf{r}, \mathbf{p}_1, \mathbf{p}_2)]$$

2

In equilibrium, $\left(\frac{\partial f}{\partial t}\right)_{\text{coll}} = 0$

2.1

Linearized collision operator

- No chemical potential

3

Odd-Even effect